

j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

LEFT rail				RIGHT rail			
+--[USB-C]--+							
I2C SDA (ADS x2 + keypad) -- 21					5V		
I2C SCL -- 22					GND ground		
DISP-L TX -> XIAO D7 -- 17				TTGO		27	DISP-L RX <- XIAO D6
(free) -- 2				T-Display		26	DISP-R RX <- XIAO D6
(free) -- 15				v1.1		25	DISP-R TX (reserved)
(free) -- 13						33	(free)
(avoid: boot/flash) -- 12				[ST7789		32	(free)
ground -- GND				TFT on		39	(input only)
3.3V -- 3V3				back]		38	(input only)
5V -- 5V						37	(input only)
ground -- GND						36	(input only)
+-----+							

on-board (no header pin): GPIO34 = battery sense, B00T = GPIO0,
GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, PCF8574 keypad @ 0x20

GPIO	Function
21	I2C SDA (ADS1115 x2, PCF8574 keypad)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, reserved)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in)